

AZURE ENVIRONMENT ASSESSMENT

Contoso

Inventory, Well-Architected findings and remediation roadmap

Report date: 29 September 2026

Inventory export date: 2026-09-20

Inventory source: sample-inventory.csv

Subscriptions in scope: 3

Resources assessed: 152

Contents

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1. Executive summary

Overall posture score: **76/100 (Fair)**

152	3	23	5
Resources	Subscriptions	Resource groups	Regions
17	73	16	78%
High findings	Medium findings	Low findings	Tag coverage

Contoso's Azure environment, comprising 152 resources across three subscriptions and five regions, has been assessed with a current overall posture score of 76/100, which is rated as Fair. This assessment reveals that while the estate demonstrates some robust practices in governance, it faces significant challenges in areas such as security, reliability, and cost optimization.

The most critical risks identified include storage accounts permitting HTTP access, allowing anonymous blob access, VMs being stopped but not deallocated, and web apps failing to enforce HTTPS. These issues pose a high risk of data exposure and unauthorized access, potentially compromising the integrity and confidentiality of sensitive information. Additionally, network security groups (NSGs) exposing management ports to the Internet and the presence of classic resources in use are also significant vulnerabilities that could be exploited by attackers.

On a more positive note, governance practices appear to be well-established, with 86% of the estate rated as Good. The tagging coverage is at 78%, ensuring that most resources have necessary metadata for effective management and compliance. Furthermore, the identified architecture patterns such as multi-region hub-and-spoke network topology, hybrid connectivity via Site-to-Site VPNs, and a federated logging approach with one workspace per subscription demonstrate a solid foundational structure.

The recommended priorities for remediation are structured according to the severity and impact of each issue. The first 30 days should focus on addressing high-risk items such as storage accounts permitting HTTP access, VMs not being deallocated properly, and web apps failing HTTPS enforcement. These actions will mitigate immediate security risks and reduce potential data breaches. Over the next 60 days, efforts should be directed towards resolving governance issues like missing required tags and ensuring that all resources are within approved regions. Finally, over the following 90 days, attention can turn to optimizing costs through measures such as managing unattached managed disks and upgrading storage replication methods.

By prioritizing these remediation steps, Contoso can significantly enhance its Azure environment's security posture, reliability, and cost efficiency, thereby protecting critical business assets and ensuring long-term operational resilience.

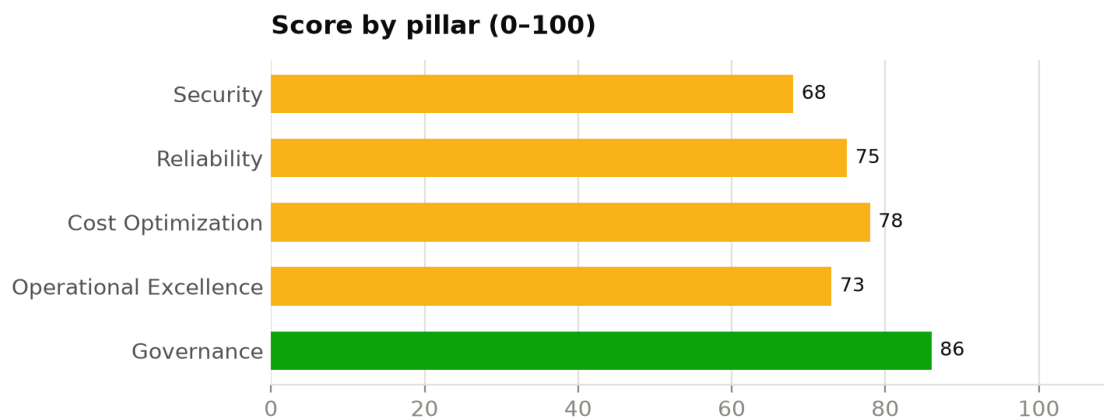
AI-drafted narrative, grounded in the assessment facts (see 2. Scope and methodology).

Key observations

- 152 resources were assessed across 3 subscription(s), 23 resource groups and 5 region(s).
- Architecture at a glance: network topology is multi-region hub-and-spoke with exceptions; hybrid connectivity is site-to-site VPN to on-premises; subscription model is landing-zone style: platform and workload subscriptions, split by environment.
- 106 finding(s) were raised: 17 high, 73 medium and 16 low severity, affecting 78 resources.
- Security is the weakest area (score 68/100); Governance is the strongest (86/100).
- 78% of resources carry all required tags (environment, owner, costCenter).
- Priority high-severity issues: Storage account permits HTTP (4); Storage account allows anonymous blob access (4); VM stopped but not deallocated (4).

Scorecard

Pillar	Score	Rating	Checks	Findings	High	Medium	Low
Security	68	Fair	51	16	12	4	0
Reliability	75	Fair	64	21	0	15	6
Cost Optimization	78	Fair	92	19	4	12	3
Operational Excellence	73	Fair	26	7	0	0	7
Governance	86	Good	305	43	1	42	0



2. Scope and methodology

The assessment is based on the resource inventory export supplied by Contoso (sample-inventory.csv). No access to the Azure environment was required. Each resource was evaluated against a rule set aligned to the Microsoft Azure Well-Architected Framework pillars (Security, Reliability, Cost Optimization, Operational Excellence) plus a Governance pillar covering tagging, regions and deployment model.

Pillar scores are the severity-weighted pass rate of all rule evaluations in that pillar (High = 5, Medium = 3, Low = 1). 80+ is rated Good, 60–79 Fair, and below 60 Poor.

Subscription	Subscription ID	Resources
Contoso-Production	11111111-1111-1111-1111-111111111111	77
Contoso-NonProduction	22222222-2222-2222-2222-222222222222	54
Contoso-Connectivity	33333333-3333-3333-3333-333333333333	21

Use of AI in this report

Narrative marked as AI-drafted was written by the open-source language model qwen2.5:7b-instruct from the verified facts produced by the assessment engine; the model was given those facts only, not the inventory itself. Every figure and rule ID in the drafted text was checked against those facts, and any passage that failed the check uses the standard wording. Scores, findings and tables are not AI-generated.

3. Architecture assessment

This section identifies the architecture patterns the environment follows, such as its network topology, hybrid connectivity and subscription model, and compares them with Microsoft's reference architectures (the Cloud Adoption Framework Azure landing zone and the Well-Architected Framework). Each area states the pattern identified, the evidence for it in the inventory, and considerations.

Confidence shows how firmly the inventory supports each conclusion: High means direct configuration evidence, Medium means inferred from resource types and naming, Low means the export lacked the data. Identity (Microsoft Entra ID), role assignments, Azure Policy and management groups are not part of a resource inventory and are outside the scope of this section.

Architecture at a glance

Area	Pattern identified	Confidence
3.1 Network topology	Multi-region hub-and-spoke with exceptions	High
3.2 Hybrid connectivity	Site-to-site VPN to on-premises	High
3.3 Traffic flow and perimeter security	Defence in depth (central firewall, WAF-protected ingress, private endpoints for PaaS, Bastion for admin access)	High
3.4 Subscription and landing zone model	Landing-zone style: platform and workload subscriptions, split by environment	Medium
3.5 Application hosting architecture	Hybrid IaaS/PaaS, three-tier (web, application, data)	Medium
3.6 Resilience and disaster recovery	Multi-region, but not in Azure region pairs; no DR replication evidence	Medium
3.7 Operations and monitoring	Federated logging (one workspace per subscription)	Medium

3.1 Network topology

Pattern identified: **Multi-region hub-and-spoke with exceptions** *(confidence: High)*

The network topology at Contoso follows a multi-region hub-and-spoke pattern, with high confidence in the evidence provided. This design centralises shared services such as Azure Bastion, Firewall, and VPN gateway within regional hubs (vnet-hub-eastus and vnet-hub-

westeurope), facilitating centralized management and security inspection. However, there are notable deviations from the Microsoft Cloud Adoption Framework's landing zone and Well-Architected Framework recommendations.

Direct peerings between spokes, such as those between vnet-prd-eastus and vnet-dev-eastus, bypass the central firewall, allowing un-inspected outbound traffic. Additionally, vnet-dev-sandbox is not peered to any network, creating an isolated environment that may expose it to unnecessary public access. These gaps could potentially undermine the security benefits of a hub-and-spoke architecture by allowing workloads in different regions and environments to communicate without central inspection or management.

To align with best practices, direct spoke-to-spoke peerings should be removed, routing all traffic through the hubs for consistent inspection. Furthermore, ensuring that vnet-dev-eastus routes 0.0.0.0/0 to the hub firewall will enforce comprehensive security measures on all outbound traffic. Lastly, a decision must be made regarding whether to integrate vnet-dev-sandbox into the main network or decommission it, as unmanaged networks can introduce additional risk and complexity.

AI-drafted narrative, grounded in the assessment facts (see 2. Scope and methodology).

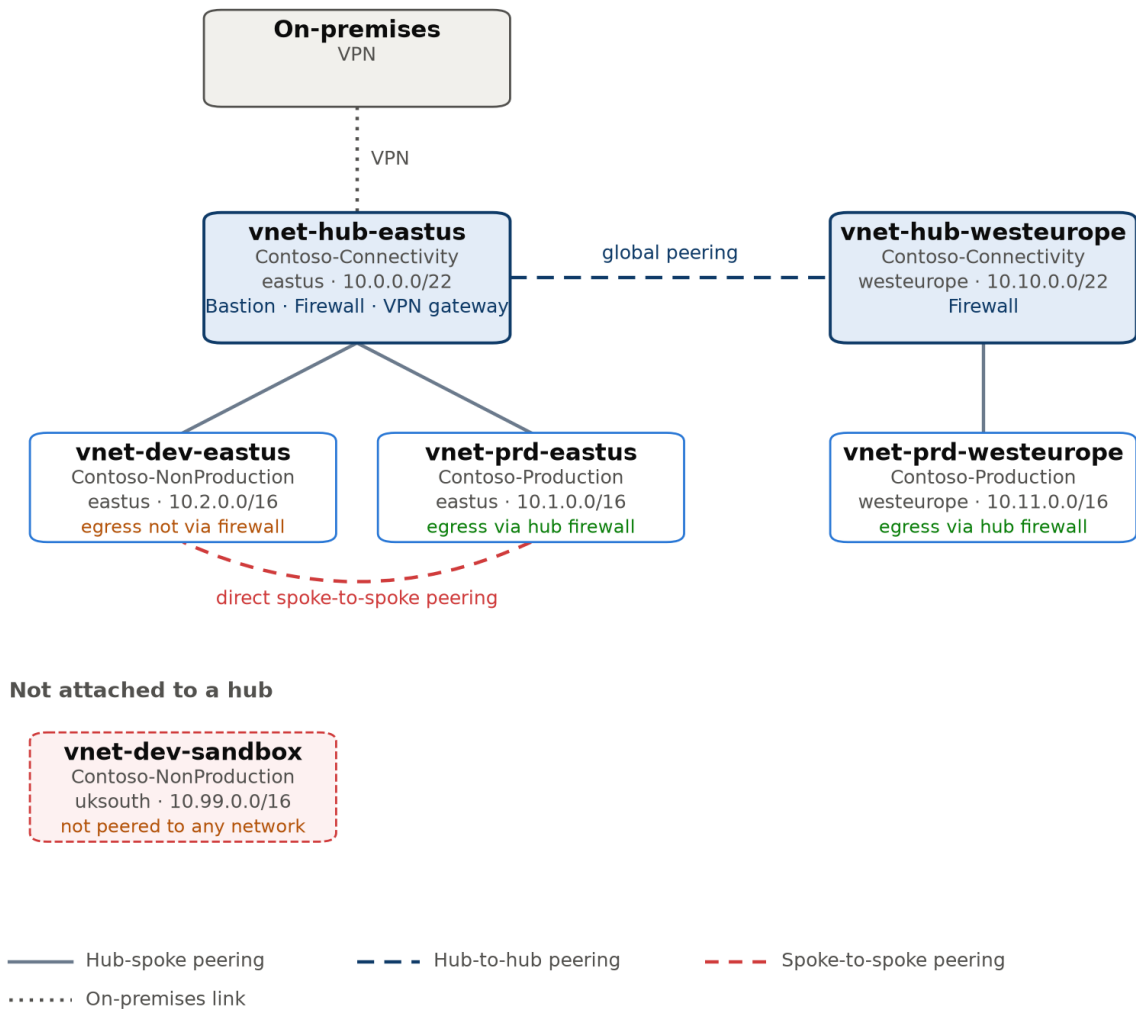


Figure: network topology derived from VNet peerings (Multi-region hub-and-spoke with exceptions).

Evidence from the inventory

- Hub vnet-hub-eastus (eastus, Contoso-Connectivity): address space 10.0.0.0/22; services: Azure Bastion, Azure Firewall and VPN gateway; spokes: vnet-prd-eastus and vnet-dev-eastus.
- Hub vnet-hub-westeurop (westeurop, Contoso-Connectivity): address space 10.10.0.0/22; services: Azure Firewall; spokes: vnet-prd-westeurop.
- Spokes vnet-prd-eastus and vnet-prd-westeurop send Internet-bound traffic to the hub firewall through a 0.0.0.0/0 route. vnet-dev-eastus has no such route, so its outbound traffic bypasses central inspection.
- vnet-prd-eastus (Contoso-Production) is peered directly with vnet-dev-eastus (Contoso-NonProduction), bypassing the hub.
- vnet-dev-sandbox (uksouth, Contoso-NonProduction) is not peered to any network.

Virtual networks

VNet	Role	Subscription	Region	Address space	Peered with	Default route via firewall
vnet-hub-eastus	Hub	Contoso-Connectivity	eastus	10.0.0.0/22	vnet-dev-eastus, vnet-hub-westeuropa, vnet-prd-eastus	-
vnet-hub-westeuropa	Hub	Contoso-Connectivity	westeuropa	10.10.0.0/22	vnet-hub-eastus, vnet-prd-westeuropa	-
vnet-dev-eastus	Spoke	Contoso-NonProduction	eastus	10.2.0.0/16	vnet-hub-eastus, vnet-prd-eastus	No
vnet-prd-eastus	Spoke	Contoso-Production	eastus	10.1.0.0/16	vnet-dev-eastus, vnet-hub-eastus	Yes
vnet-prd-westeuropa	Spoke	Contoso-Production	westeuropa	10.11.0.0/16	vnet-hub-westeuropa	Yes
vnet-dev-sandbox	Isolated	Contoso-NonProduction	uksouth	10.99.0.0/16	-	No

Considerations against the reference architecture

- Remove direct spoke-to-spoke peerings and route that traffic through the hub firewall, particularly between production and non-production (Contoso-NonProduction and Contoso-Production).
- Add a user-defined route (0.0.0.0/0 to the hub firewall) to vnet-dev-eastus so all egress is inspected.
- Decide whether vnet-dev-sandbox should join the hub or be removed; unmanaged islands tend to accumulate public exposure.

Related findings

- SEC-007 NSG exposes management ports to the Internet: **High**, 2 resources (see 6.1 Security)

3.2 Hybrid connectivity

Pattern identified: **Site-to-site VPN to on-premises** (*confidence: High*)

The current architecture employs a site-to-site VPN to connect on-premises infrastructure, adhering to the recommended shared-gateway design. The evidence is high, as indicated by the presence of vgw-hub-eastus, which is configured with a VpnGw2AZ SKU and operates in an active-standby mode within vnet-hub-eastus. This setup ensures redundancy and reliability for the connection.

However, there are gaps when compared to the Microsoft reference architecture. The current design only includes a gateway in the eastus region, while spokes in westeuropa rely on cross-region connectivity through the hub gateway. If the eastus region becomes unavailable, this could disrupt on-premises connectivity. To mitigate this risk, it is recommended to deploy additional gateways in each regional hub, ensuring that all regions can independently establish a connection to on-premises sites.

Additionally, the current setup relies solely on IPsec tunnels over the Internet for production workloads, which may introduce latency and bandwidth limitations. For critical applications,

leveraging ExpressRoute with a VPN as a backup could provide more robust connectivity options, enhancing both performance and reliability. Configuring the gateways in an active-active mode would also reduce the risk of single points of failure by ensuring that two on-premises devices can maintain the connection simultaneously.

In practice, these gaps imply potential vulnerabilities to regional outages and limitations in network performance for mission-critical workloads. Addressing these issues through additional gateway deployments and enhanced connectivity options will strengthen the overall hybrid environment's resilience and efficiency.

AI-drafted narrative, grounded in the assessment facts (see 2. Scope and methodology).

Evidence from the inventory

- vgw-hub-eastus: Vpn gateway, SKU VpnGw2AZ, zone-redundant, active-standby, deployed in vnet-hub-eastus.
- lng-onprem-dc1: on-premises site at 203.0.113.10 advertising 192.168.0.0/16.
- cn-hub-eastus-to-dc1: IPsec connection, status Connected.

Considerations against the reference architecture

- Only eastus has a gateway; spokes in westeurope reach on-premises across regions, and lose on-premises connectivity if that region is unavailable. Add a gateway in each regional hub.
- A single VPN tunnel runs over the Internet; for production workloads with latency or bandwidth needs, consider ExpressRoute with VPN as backup.
- Configure VPN gateways as active-active with two on-premises devices to remove a single point of failure.

3.3 Traffic flow and perimeter security

Pattern identified: Defence in depth (central firewall, WAF-protected ingress, private endpoints for PaaS, Bastion for admin access) *(confidence: High)*

The identified pattern of Defence in Depth for traffic flow and perimeter security at Contoso closely aligns with the principles outlined in both the Cloud Adoption Framework (CAF) Landing Zone and the Well-Architected Framework. The central firewall, WAF-protected ingress, private endpoints for PaaS services, and Bastion for administrative access demonstrate a robust approach to securing network traffic.

However, there are notable gaps that indicate areas for improvement. Specifically, some Network Security Groups (NSGs) still allow RDP or SSH from the Internet, which contradicts the use of Azure Bastion. This leaves potential vulnerabilities in place, as Internet-sourced connections bypass the intended security measures. Additionally, while 15 PaaS services accept public network traffic, only two currently utilise private endpoints, indicating a need for broader implementation.

To align more closely with best practices, it is recommended to remove any existing Internet-sourced RDP/SSH rules and extend private endpoint usage across all relevant services. Implementing DDoS Network Protection where appropriate will further enhance

the security posture of internet-facing workloads. These actions are critical to ensuring a comprehensive Defence in Depth strategy that minimises exposure to potential threats, thereby protecting both data integrity and system availability.

AI-drafted narrative, grounded in the assessment facts (see 2. Scope and methodology).

Evidence from the inventory

- Web application firewall: agw-prd-eastus.
- DDoS Network Protection plan: none.

Considerations against the reference architecture

- Remove Internet-sourced RDP/SSH rules now that Bastion is available.
- Extend private endpoints to the remaining PaaS services and disable their public network access (stprddata00, stprddata01, stprddata02, stprddata03, stprddata04 and kv-prd-00 ...).
- Consider DDoS Network Protection for virtual networks with Internet-facing workloads.

Related findings

- SEC-007 NSG exposes management ports to the Internet: **High**, 2 resources (see 6.1 Security)
- SEC-005 SQL server reachable from public networks: **Medium**, 1 resource (see 6.1 Security)
- SEC-002 Storage account allows anonymous blob access: **High**, 4 resources (see 6.1 Security)
- SEC-006 Web app does not enforce HTTPS: **High**, 2 resources (see 6.1 Security)

3.4 Subscription and landing zone model

Pattern identified: **Landing-zone style: platform and workload subscriptions, split by environment** (*confidence: Medium*)

The identified subscription and landing zone model follows a platform and workload separation, with three distinct subscriptions: Contoso-Connectivity (platform), Contoso-NonProduction (workload), and Contoso-Production (workload). This pattern is consistent with the Cloud Adoption Framework's recommendation for separating shared services from application workloads. However, there are notable gaps compared to the Microsoft reference architecture.

Firstly, the absence of a dedicated management subscription means that central logging, automation, and backup functionalities are dispersed across multiple subscriptions—specifically, Log Analytics workspaces reside in both non-production and production environments. This could lead to inconsistencies in monitoring and governance practices, as these functions should ideally be centralized for easier management and compliance.

Secondly, there is no identified identity subscription, which suggests that domain controllers or other identity services may not be properly isolated. According to the

reference architecture, isolating such critical components in a dedicated subscription enhances security and simplifies administrative tasks.

Lastly, while the subscriptions are structured according to the landing zone model, management groups, Azure Policy assignments, and role-based access controls (RBAC) are not documented within this inventory. This oversight means that governance and compliance checks need to be conducted separately to ensure alignment with best practices outlined in the Well-Architected Framework.

These gaps imply potential challenges in maintaining a cohesive and secure environment, particularly around centralized management and identity services. Addressing these areas will help Contoso achieve a more robust and compliant Azure architecture.

AI-drafted narrative, grounded in the assessment facts (see 2. Scope and methodology).

Subscription roles

Subscription	Inferred role	Resources	Main resource types
Contoso-Connectivity	Platform: connectivity	21	Public IP addresses (6), Network security groups (4), Virtual networks (2)
Contoso-NonProduction	Workload: non-production	54	Managed disks (14), Virtual machines (10), Network interfaces (10)
Contoso-Production	Workload: production	77	Managed disks (20), Virtual machines (14), Network interfaces (14)

Considerations against the reference architecture

- There is no dedicated management subscription; Log Analytics workspaces sit in Contoso-NonProduction and Contoso-Production. The landing zone model places central logging, automation and backup in a management subscription.
- No identity subscription was found. If domain controllers or identity services run in Azure, isolate them in a dedicated identity subscription.
- Management groups, Azure Policy assignments and RBAC are not part of a resource inventory; review them separately to confirm the landing zone governance.

3.5 Application hosting architecture

Pattern identified: Hybrid IaaS/PaaS, three-tier (web, application, data)

(confidence: Medium)

The application hosting architecture at Contoso employs a hybrid Infrastructure as a Service (IaaS) and Platform as a Service (PaaS) model, with a three-tier design comprising the presentation, application, and data tiers. This pattern is identified with medium confidence based on the inventory evidence provided.

Comparing this to Microsoft's Cloud Adoption Framework landing zone and Well-Architected Framework, we observe that while the current architecture effectively separates concerns into distinct tiers, there are opportunities for improvement. Specifically, the absence of VM scale sets means that stateless components such as web apps do not benefit from automatic scaling capabilities, which could lead to underutilization or over-provisioning

during peak loads. Additionally, several workloads hosted on IaaS virtual machines present operational overhead in terms of patch management and availability maintenance. Reevaluating these components for potential replatforming to PaaS services like App Service, Azure SQL Database, or Kubernetes (AKS) could streamline operations and reduce the burden on IT staff.

In practice, failing to leverage scale sets and other managed services may result in suboptimal resource utilization and increased operational costs. For instance, the identified gaps in VM sizing, deallocation of stopped instances, and consolidation of App Service plans highlight areas where automation and optimization can be further enhanced. Addressing these issues will not only improve cost efficiency but also align more closely with Microsoft's best practices for a robust and scalable cloud architecture.

AI-drafted narrative, grounded in the assessment facts (see 2. Scope and methodology).

Evidence from the inventory

- Virtual machines: 24; scale sets: 0; AKS clusters: 0; web apps: 8; function apps: 0; databases and caches: 8.

Considerations against the reference architecture

- No VM scale sets are used, so VM capacity does not scale automatically. Stateless tiers are good candidates for scale sets or App Service.
- Review VM-hosted components for replatforming to PaaS (App Service, Azure SQL, AKS) to reduce operational overhead.

Related findings

- OPS-001 Previous-generation VM size: **Low**, 5 resources (see 6.4 Operational Excellence)
- COST-003 VM stopped but not deallocated: **High**, 4 resources (see 6.3 Cost Optimization)
- COST-005 Empty App Service plan: **Medium**, 2 resources (see 6.3 Cost Optimization)

3.6 Resilience and disaster recovery

Pattern identified: Multi-region, but not in Azure region pairs; no DR replication evidence *(confidence: Medium)*

The resilience and disaster recovery (RDR) pattern identified in the environment is multi-region, but not within Azure region pairs. This configuration provides a basic level of redundancy across four regions—eastus, westeurope, uksouth, and centralindia—but lacks data replication mechanisms necessary for effective disaster recovery. The confidence in this assessment is medium based on the available evidence.

Compared to the Microsoft reference architecture outlined by the Cloud Adoption Framework (CAF) landing zone and Well-Architected Framework, several gaps are evident. Firstly, there is no deployment of Recovery Services vaults or Azure Backup solutions, which would be critical for protecting VMs and databases. Secondly, while some workloads use

availability zones, this is not consistently applied across all resources, and there is no defined disaster recovery strategy or tested failover plans.

In practice, these gaps mean that the environment's ability to recover from regional outages or disasters is limited. Without proper data replication and backup strategies, significant downtime could occur if a region fails. The absence of Recovery Services vaults also leaves critical workloads unprotected against data loss. Implementing these recommendations would enhance the overall resilience and ensure business continuity in the event of unforeseen disruptions.

AI-drafted narrative, grounded in the assessment facts (see 2. Scope and methodology).

Considerations against the reference architecture

- Deploy Recovery Services vaults and back up VMs and databases; define RPO/RTO per workload.
- Decide the DR strategy per workload (active/active, active/passive or backup-and-restore), then replicate data and test failover.
- Place production VMs across availability zones (or use zone-redundant scale sets) to survive a datacenter failure.

Related findings

- REL-001 VM has no zone or availability set: **Medium**, 15 resources (see 6.2 Reliability)
- REL-003 Storage account uses locally redundant replication: **Low**, 3 resources (see 6.2 Reliability)
- REL-004 SQL database not zone redundant: **Low**, 3 resources (see 6.2 Reliability)

3.7 Operations and monitoring

Pattern identified: Federated logging (one workspace per subscription)

(confidence: Medium)

The identified pattern of federated logging, with one workspace per subscription (log-prd-01 in Contoso-Production and log-dev-01 in Contoso-NonProduction), simplifies access control by environment but complicates cross-environment monitoring. This approach aligns partially with the Cloud Adoption Framework's landing zone model, which recommends a central workspace in the management subscription for centralized visibility. However, it diverges from this recommendation as each workload environment maintains its own workspace. The absence of a dedicated workspace for Contoso-Connectivity further exacerbates this gap.

The current setup makes it challenging to achieve a unified security view and cross-environment analytics, which are critical for comprehensive operational monitoring and incident response. This could lead to fragmented insights and increased complexity in managing security and compliance across environments. The low severity of the OPS-002 finding related to Log Analytics retention underscores the importance of addressing these gaps to ensure that all resources meet audit and investigative requirements effectively.

AI-drafted narrative, grounded in the assessment facts (see 2. Scope and methodology).

Considerations against the reference architecture

- The landing zone model recommends a central workspace in the management subscription, with resource-context access for workload teams.

Related findings

- OPS-002 Log Analytics retention below target: **Low**, 2 resources (see 6.4 Operational Excellence)

4. Infrastructure overview

This section describes the environment as understood from the inventory export: what it contains, how it is organised and how each area is configured. Each part ends with the findings that relate to it, so the description can be read alongside the findings in section 6 and the roadmap in section 7.

4.1 Estate at a glance

The estate comprises 152 Azure resources distributed across three subscriptions and 23 resource groups, with deployments in five regions. The primary region, eastus, hosts 49% of the total resources, followed by westeurope (27%), uksouth (18%), and a smaller presence in centralindia (5%). This regional distribution suggests a balanced approach to ensure geographical redundancy and local compliance requirements.

From a functional perspective, compute and networking dominate at 38% each, indicating a focus on scalable infrastructure. The significant number of managed disks (34) and virtual machines (24) supports this, highlighting the use of IaaS for core operations. However, the presence of classic resources (1%) and legacy application services (8%) could pose maintenance challenges and security risks, as these are generally less secure and harder to manage compared to modern Azure services.

The high proportion of managed disks and virtual machines also implies a robust infrastructure capable of handling significant workloads, which is beneficial for operational resilience. However, the mixed hosting model, with both IaaS and some classic resources, may introduce complexity in management and could benefit from consolidation into more modern, secure services to streamline operations and enhance security posture.

AI-drafted narrative, grounded in the assessment facts (see 2. Scope and methodology).

Resources by functional area

Area	Resources	% of estate	Main resource types
Compute	58	38%	Managed disks (34), Virtual machines (24)
Networking	57	38%	Network interfaces (24), Network security groups (7), Virtual networks (6)
Application hosting	12	8%	App Services (8), App Service plans (4)
Storage	10	7%	Storage accounts (10)
Databases	8	5%	SQL databases (6), SQL servers (2)
Security services	4	3%	Key vaults (4)

Area	Resources	% of estate	Main resource types
Monitoring and management	2	1%	Log Analytics workspaces (2)
Classic (legacy) resources	1	1%	Classic cloud services (1)

4.2 Subscriptions and resource organisation

The infrastructure is organised into three distinct Azure subscriptions: Contoso-Production, Contoso-NonProduction, and Contoso-Connectivity. This separation helps in managing different environments—production, development, and connectivity-related resources—separately. On average, each resource group contains 6.6 resources, with the largest groups being rg-prd-app01 (19), rg-dev-app01 (13), and rg-prd-app02 (12).

The Contoso-Production subscription houses 77 resources across nine resource groups in four regions, primarily focusing on managed disks, virtual machines, and network interfaces. This setup ensures that critical production workloads are well-segregated from development environments, enhancing security and operational resilience. Similarly, the Contoso-NonProduction environment contains 54 resources, mainly comprising similar types of infrastructure but tagged for development purposes.

The Contoso-Connectivity subscription, with its 21 resources across four resource groups in three regions, manages network-related services such as public IP addresses, network security groups, and virtual networks. This separation is crucial for maintaining a clear distinction between application and connectivity resources, which can improve overall network security and management efficiency.

This structured organisation supports better governance, easier compliance checks, and improved operational visibility, ensuring that the business can manage costs more effectively while maintaining robust security and resilience across its environments.

AI-drafted narrative, grounded in the assessment facts (see 2. Scope and methodology).

- **Contoso-Production:** 77 resources in 9 resource groups across 4 regions (centralindia, eastus, uksouth and westeurope). It mainly contains managed disks (20), virtual machines (14) and network interfaces (14). Resources are tagged as environment 'prod'.
- **Contoso-NonProduction:** 54 resources in 10 resource groups across 4 regions (centralindia, eastus, uksouth and westeurope). It mainly contains managed disks (14), virtual machines (10) and network interfaces (10). Resources are tagged as environment 'dev'.
- **Contoso-Connectivity:** 21 resources in 4 resource groups across 3 regions (eastus, global and westeurope). It mainly contains public IP addresses (6), network security groups (4) and virtual networks (2). Resources are tagged as environment 'shared'.

Subscription summary

Subscription	Resources	Resource groups	Regions	Main resource types
Contoso-	77	9	centralindia,	Managed disks (20), Virtual

Subscription	Resources	Resource groups	Regions	Main resource types
Production			eastus, uksouth, westeurope	machines (14), Network interfaces (14)
Contoso-NonProduction	54	10	centralindia, eastus, uksouth, westeurope	Managed disks (14), Virtual machines (10), Network interfaces (10)
Contoso-Connectivity	21	4	eastus, global, westeurope	Public IP addresses (6), Network security groups (4), Virtual networks (2)

4.3 Regional footprint

The regional footprint of the environment spans five regions: eastus, westeurope, uksouth, centralindia, and a global region. While deploying resources across multiple regions provides an opportunity for regional resilience, only three of these—eastus, westeurope, and uksouth—are officially approved. The remaining 8 out-of-policy resources are located in the centralindia region.

This configuration introduces both benefits and challenges. From a security perspective, having resources distributed across different geopolitical boundaries can enhance compliance with local data sovereignty regulations. However, it also increases complexity in managing and securing these resources independently. For resilience, deploying workloads only in approved regions means that regional failures could impact operations if not properly failover-capable.

The presence of 8 out-of-policy resources (GOV-002) highlights a potential risk to governance and operational efficiency. These resources may be subject to additional regulatory scrutiny or compliance issues, which could affect the overall security posture. Additionally, maintaining these non-compliant regions incurs unnecessary costs and complicates management efforts.

To mitigate these risks, it is recommended to apply the 'Allowed locations' policy as per GOV-002 and plan for the migration of out-of-policy resources into approved regions. This will streamline governance, enhance security, and potentially reduce operational overhead.

AI-drafted narrative, grounded in the assessment facts (see 2. Scope and methodology).

Resources by region

Region	Resources	% of estate	Main areas
eastus	74	49%	Networking (32), Compute (25), Application hosting (7)
westeurope	41	27%	Compute (14), Networking (14), Storage (5)
uksouth	27	18%	Compute (15), Networking (8), Databases (2)
centralindia	8	5%	Compute (4), Storage (2), Networking (1)
global	2	1%	Networking (2)

Related findings

- GOV-002 Resource outside approved regions: **Medium**, 8 resources (see 6.5 Governance)

4.4 Compute

The compute environment consists of 24 virtual machines (VMs) distributed across two Azure subscriptions and four regions, with a mix of Linux (20 VMs) and Windows (4 VMs). The majority of the VM sizes are D-series v5 (13), E-series v5 (4), and A-series v2 (3), with specific common sizes including Standard_D2s_v5 (9), Standard_E8s_v5 (4), and Standard_D4s_v5 (4).

From a security and resilience perspective, 79% of the VMs are currently running, while one is deallocated and four are stopped but still allocated. This means that these stopped VMs continue to incur costs, which could be optimized by deallocating them or scheduling an auto-shutdown as per COST-003 and COST-004 recommendations.

Resilience is another critical aspect; 9 VMs are pinned to availability zones, providing a level of fault tolerance. However, the remaining 15 VMs lack both zone and availability-set protection, meaning they depend on a single host or datacenter for redundancy. This configuration could pose a risk in case of regional outages (REL-001). Deploying these VMs across multiple zones or into an availability set would enhance resilience.

Cost optimization is also a key area; 15 VMs are not utilizing the latest generation SKUs, which may offer better price-performance (OPS-001). Additionally, there are 10 unattached managed disks that are still billed but unused, suggesting potential savings through deletion or snapshotting and subsequent removal as recommended by COST-001.

Overall, while the current setup meets basic operational needs, improvements in resilience, cost management, and VM sizing can significantly enhance both security and efficiency.

AI-drafted narrative, grounded in the assessment facts (see 2. Scope and methodology).

Virtual machines by size

VM size	Series	Count
Standard_D2s_v5	D-series v5	9
Standard_E8s_v5	E-series v5	4
Standard_D4s_v5	D-series v5	4
Standard_A2_v2	A-series v2	3
Standard_DS2_v2	DS-series v2	2
Standard_B2ms	B-series	2

Related findings

- REL-001 VM has no zone or availability set: **Medium**, 15 resources (see 6.2 Reliability)
- COST-003 VM stopped but not deallocated: **High**, 4 resources (see 6.3 Cost Optimization)
- COST-004 Deallocated VM: **Low**, 1 resource (see 6.3 Cost Optimization)

- COST-001 Unattached managed disk: **Medium**, 10 resources (see 6.3 Cost Optimization)
- OPS-001 Previous-generation VM size: **Low**, 5 resources (see 6.4 Operational Excellence)

4.5 Networking

The networking infrastructure at Contoso comprises 57 resources, including network interfaces, security groups, virtual networks, and load balancers. This setup supports a distributed environment with six virtual networks across regions such as eastus, uksouth, and westeurope, bolstered by two Azure Firewalls for enhanced security. Traffic filtering is managed through seven Network Security Groups (NSGs) that collectively apply 9 custom rules, though some NSGs inadvertently expose management ports to the Internet, posing a significant security risk (SEC-007). Additionally, six public IP addresses are allocated, with two remaining unassociated and thus unused, indicating potential cost inefficiencies (COST-002).

From a security perspective, exposing management ports to the Internet can lead to unauthorized access attempts. Implementing Azure Bastion or Just-in-Time VM access would mitigate this risk by restricting direct internet access to critical services. Regarding resilience, while the current setup is robust, ensuring proper network topology and connectivity between virtual networks will be crucial for maintaining service availability across regions.

Cost-wise, releasing unassociated public IP addresses can help Contoso reduce unnecessary expenses without impacting operational needs. Overall, optimizing these networking elements will enhance both security and cost efficiency, aligning with broader IT strategy goals.

AI-drafted narrative, grounded in the assessment facts (see 2. Scope and methodology).

Network resources

Resource type	Count	Regions
Network interfaces	24	centralindia, eastus, uksouth, westeurope
Network security groups	7	eastus, westeurope
Virtual networks	6	eastus, uksouth, westeurope
Public IP addresses	6	eastus, westeurope
Azure Firewalls	2	eastus, westeurope
routetables	2	eastus, westeurope
privateendpoints	2	eastus
privatednszones	2	global
virtualnetworkgateways	1	eastus
localnetworkgateways	1	eastus
connections	1	eastus
bastionhosts	1	eastus
Application gateways	1	eastus
Load balancers	1	eastus

Related findings

- SEC-007 NSG exposes management ports to the Internet: **High**, 2 resources (see 6.1 Security)
- COST-002 Unassociated public IP address: **Low**, 2 resources (see 6.3 Cost Optimization)

4.6 Storage

The infrastructure comprises 10 storage accounts, all of the StorageV2 type. Replication strategies vary: four use GRS (Geo-redundant replication), three employ LRS (Locally redundant replication), and another three utilise ZRS (Zone-redundant replication). This setup ensures that data is replicated within a single region for local redundancy, while also providing an additional copy in a paired region with GRS. ZRS offers enhanced resilience by replicating data across availability zones.

From a security perspective, four accounts permit HTTP traffic, which could expose sensitive data to potential threats (SEC-001). Enforcing HTTPS is recommended to ensure secure data transfer. Additionally, four storage accounts allow anonymous blob access, potentially compromising data integrity and confidentiality (SEC-002), and two accept TLS versions lower than 1.2, leaving them vulnerable to certain types of cyberattacks (SEC-003). Implementing TLS 1.2 or higher is advised.

Regarding reliability, three out of ten storage accounts use LRS, which offers minimal resilience within a single datacenter (REL-003). For critical production data that requires regional redundancy and availability, switching to ZRS or GZRS would be beneficial. This change aligns with the need for enhanced disaster recovery capabilities and reduced risk of data loss due to localized outages.

AI-drafted narrative, grounded in the assessment facts (see 2. Scope and methodology).

Storage accounts

Account	Subscription	Replication	HTTPS only	Min TLS	Anonymous blob access
stdevdata00	Contoso-NonProduction	Standard_GRS	Yes	TLS1_2	Allowed
stdevdata01	Contoso-NonProduction	Standard_GRS	No	TLS1_2	Allowed
stdevdata02	Contoso-NonProduction	Standard_LRS	Yes	TLS1_0	Blocked
stdevdata03	Contoso-NonProduction	Standard_ZRS	No	TLS1_2	Blocked
stdevdata04	Contoso-NonProduction	Standard_GRS	Yes	TLS1_0	Blocked
stprddata00	Contoso-Production	Standard_LRS	Yes	TLS1_2	Blocked
stprddata01	Contoso-Production	Standard_GRS	Yes	TLS1_2	Blocked
stprddata02	Contoso-Production	Standard_ZRS	Yes	TLS1_2	Allowed
stprddata03	Contoso-Production	Standard_ZRS	No	TLS1_2	Blocked
stprddata04	Contoso-Production	Standard_LRS	No	TLS1_2	Allowed

Related findings

- SEC-001 Storage account permits HTTP: **High**, 4 resources (see 6.1 Security)
- SEC-002 Storage account allows anonymous blob access: **High**, 4 resources (see 6.1 Security)
- SEC-003 Storage account accepts TLS < 1.2: **Medium**, 2 resources (see 6.1 Security)
- REL-003 Storage account uses locally redundant replication: **Low**, 3 resources (see 6.2 Reliability)

4.7 Databases

The database environment at Contoso currently employs Azure SQL, with 2 logical servers hosting a total of six databases across different service tiers: S1 (3), Basic (2), and GP_Gen5_2 (1). This setup indicates a mix of performance levels, where the S1 tier is likely used for more transactional workloads, while the Basic tier may be allocated to less demanding tasks. The single database on the GP_Gen5_2 tier suggests it might handle higher-performance requirements.

From a security perspective, one logical server (sql-dev-01) allows public network access, which poses a medium-severity risk as identified by SEC-005. This configuration could expose these databases to potential unauthorized access or attacks from the internet. The recommendation is to disable public network access and connect through Private Endpoints for enhanced security.

In terms of resilience, only one out of six databases (REL-004) benefits from zone redundancy, indicating a low-severity risk. Zone redundancy ensures that if an Azure region fails, data can still be accessed from another zone within the same region. Given this finding, it is advisable to enable zone redundancy on all business-critical databases to improve overall reliability and avoid potential downtime.

AI-drafted narrative, grounded in the assessment facts (see 2. Scope and methodology).

Related findings

- SEC-005 SQL server reachable from public networks: **Medium**, 1 resource (see 6.1 Security)
- REL-004 SQL database not zone redundant: **Low**, 3 resources (see 6.2 Reliability)

4.8 Application hosting

Application hosting in the environment is configured to use Azure App Service, with eight web applications distributed across four different pricing tiers: two P1v3 plans, one S1 plan, and one B1 plan. Six of these apps enforce HTTPS-only access, enhancing security by preventing data interception over HTTP connections.

However, there are implications for both security and cost management. The enforcement of HTTPS on six out of eight applications indicates a proactive stance towards securing web traffic, which is crucial to protect sensitive information from eavesdropping. Nonetheless, two App Service plans (asp-dev-00 and asp-dev-01) host no active apps but still incur charges, highlighting inefficiencies in resource utilization.

According to the findings, SEC-006 identifies that two of eight evaluated resources do not enforce HTTPS-only access, suggesting a potential vulnerability. COST-005 points out that these empty App Service plans are unnecessary and should be deleted or consolidated with active plans (Rule ID: COST-005), which could lead to significant cost savings without compromising functionality.

AI-drafted narrative, grounded in the assessment facts (see 2. Scope and methodology).

Related findings

- SEC-006 Web app does not enforce HTTPS: **High**, 2 resources (see 6.1 Security)
- COST-005 Empty App Service plan: **Medium**, 2 resources (see 6.3 Cost Optimization)

4.9 Security services

The security services in the environment are managed through four Key Vaults: kv-dev-00, kv-dev-01, kv-prd-00 and kv-prd-01. Three out of these four vaults have purge protection enabled, which ensures that deleted secrets cannot be permanently removed during a specified retention period. This setup enhances the security posture by providing an additional safeguard against both malicious and accidental deletions.

While this configuration is effective for three of the Key Vaults, one remains unprotected as per SEC-004 (Security, Medium severity). The lack of purge protection in kv-dev-01 poses a risk, as it could lead to irreversible data loss if secrets are mistakenly or intentionally deleted. Enabling purge protection across all vaults would mitigate this risk and align with best security practices.

The use of two Azure Firewalls—one each in the eastus and westeurope regions—provides robust network perimeter protection. This dual-region deployment ensures that even if one firewall is compromised, the other can still protect the environment, thereby enhancing resilience. However, it's important to ensure continuous monitoring and maintenance of both firewalls to maintain this level of security and resilience.

AI-drafted narrative, grounded in the assessment facts (see 2. Scope and methodology).

Related findings

- SEC-004 Key Vault purge protection disabled: **Medium**, 1 resource (see 6.1 Security)

4.10 Monitoring and management

The monitoring and management setup in the environment involves collecting logs across two Log Analytics workspaces, log-dev-01 and log-prd-01, with a 30-day retention period. This configuration ensures that operational data is retained for a sufficient duration to facilitate troubleshooting and compliance checks. However, the absence of an Azure Recovery Services vault indicates no current backup strategy for virtual machines or databases, which could pose risks in case of data loss or system failures.

The short log retention period (30 days) identified in OPS-002 highlights that while operational excellence is being maintained to some extent, it may not fully meet the organization's audit and investigative requirements. This could lead to potential gaps in

forensic analysis during security incidents or regulatory compliance checks. To mitigate this risk, increasing log retention or implementing a more comprehensive archiving strategy would be advisable.

In summary, while the current setup supports basic monitoring needs, enhancing both the logging duration and backup strategies is crucial for improving overall resilience and ensuring data integrity.

AI-drafted narrative, grounded in the assessment facts (see 2. Scope and methodology).

Related findings

- OPS-002 Log Analytics retention below target: **Low**, 2 resources (see 6.4 Operational Excellence)

4.11 Governance and tagging

In the governance and tagging section, 89% of resources are tagged with at least one tag, and 78% carry all required tags—environment, owner, and costCenter. The use of three distinct keys (costCenter, environment, and owner) ensures a structured approach to resource management. However, there is room for improvement; GOV-001 indicates that 34 resources are missing at least one required tag, suggesting potential gaps in accountability and visibility. Additionally, the presence of a single classic deployment model (legacy-cloudservice) poses a high-risk issue as it uses the retired Azure Service Manager model, which needs to be migrated to Azure Resource Manager before platform retirement.

These findings imply that while tagging is largely compliant, there are critical areas for enhancement. Enforcing tagging standards through Azure Policy will help mitigate risks associated with missing or incomplete tags. The classic resource also highlights a need for proactive migration strategies to ensure ongoing system resilience and compliance.

AI-drafted narrative, grounded in the assessment facts (see 2. Scope and methodology).

Required tag coverage

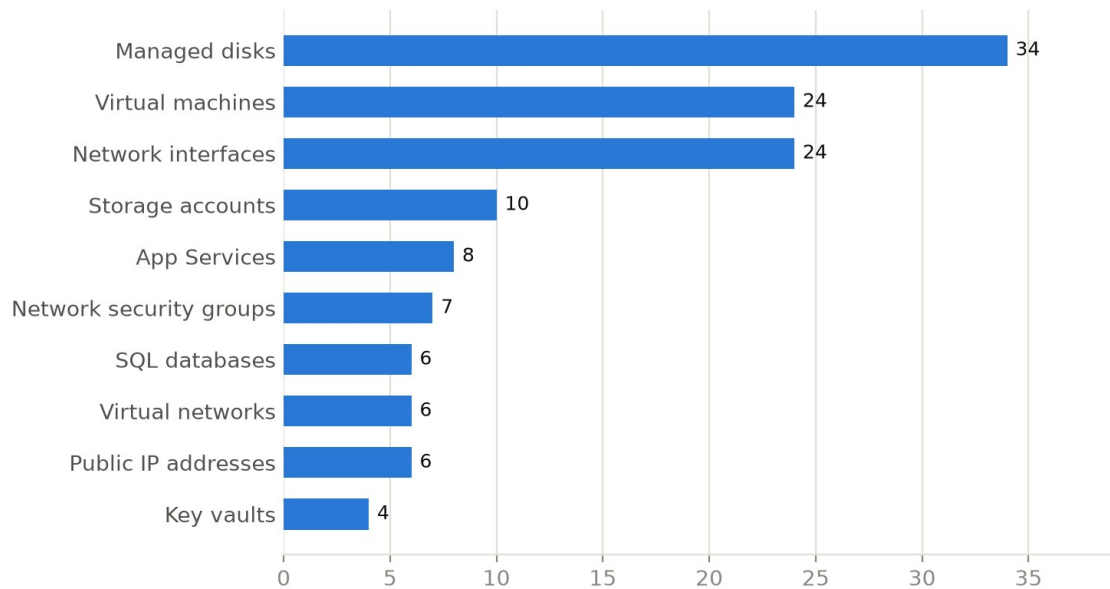
Tag	Resources tagged	Coverage	Most common values
environment	125	82%	prod (63), dev (44), shared (18)
owner	125	82%	app-team (44), platform (44), data-team (37)
costCenter	126	83%	CC300 (43), CC200 (42), CC100 (41)

Related findings

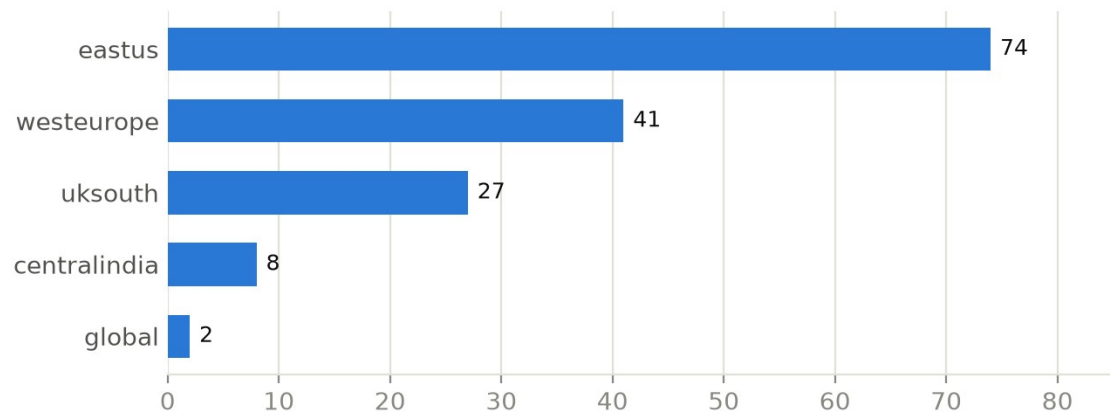
- GOV-001 Required tags missing: **Medium**, 34 resources (see 6.5 Governance)
- GOV-002 Resource outside approved regions: **Medium**, 8 resources (see 6.5 Governance)
- GOV-003 Classic (ASM) resource in use: **High**, 1 resource (see 6.5 Governance)

5. Inventory charts and statistics

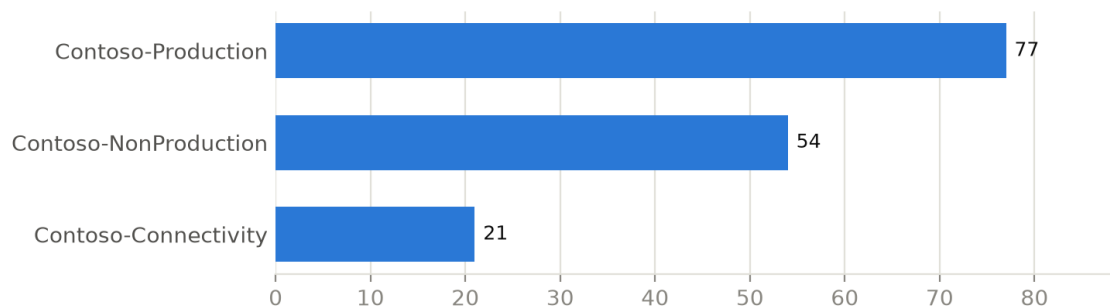
Top resource types



Resources by region



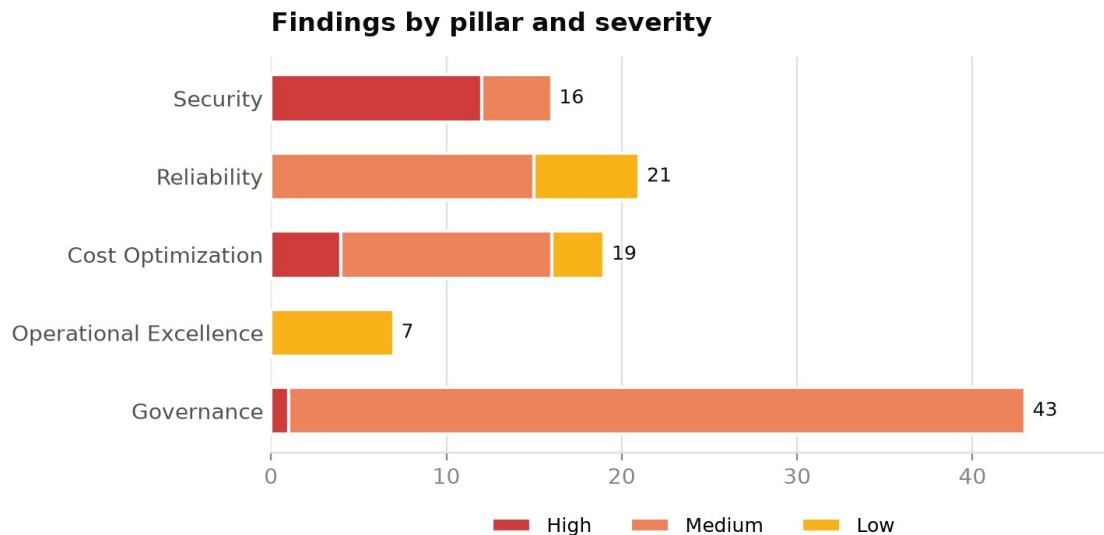
Resources by subscription



Resource types

Resource type	Count	% of estate
Managed disks	34	22.4%
Virtual machines	24	15.8%
Network interfaces	24	15.8%
Storage accounts	10	6.6%
App Services	8	5.3%
Network security groups	7	4.6%
SQL databases	6	3.9%
Virtual networks	6	3.9%
Public IP addresses	6	3.9%
Key vaults	4	2.6%
App Service plans	4	2.6%
SQL servers	2	1.3%
Log Analytics workspaces	2	1.3%
Azure Firewalls	2	1.3%
routetables	2	1.3%
privateendpoints	2	1.3%
privatednszones	2	1.3%
Classic cloud services	1	0.7%
virtualnetworkgateways	1	0.7%
localnetworkgateways	1	0.7%
connections	1	0.7%
bastionhosts	1	0.7%
Application gateways	1	0.7%
Load balancers	1	0.7%

6. Assessment findings



6.1 Security — 68/100 (Fair)

SEC-001: Storage account permits HTTP

Severity: High Affected: 4 of 10 evaluated resources

Recommendation: Enable 'Secure transfer required' on the storage account.

Examples: stdevdata01, stdevdata03, stprddata03, stprddata04

Context: see 4.6 Storage in the infrastructure overview.

SEC-002: Storage account allows anonymous blob access

Severity: **High** Affected: 4 of 10 evaluated resources

Recommendation: Set allowBlobPublicAccess to false and use SAS or Entra ID for access.

Examples: stdevdata00, stdevdata01, stprddata02, stprddata04

Context: see 4.6 Storage in the infrastructure overview.

SEC-006: Web app does not enforce HTTPS

Severity: **High** Affected: 2 of 8 evaluated resources

Recommendation: Enable 'HTTPS Only' on the App Service.

Examples: app-dev-01, app-dev-02

Context: see 4.8 Application hosting in the infrastructure overview.

SEC-007: NSG exposes management ports to the Internet

Severity: **High** Affected: 2 of 7 evaluated resources

Recommendation: Remove Internet-sourced RDP/SSH rules; use Azure Bastion or Just-in-Time VM access.

Examples: nsg-mgmt-eastus, nsg-mgmt-west europe

Context: see 4.5 Networking in the infrastructure overview.

SEC-003: Storage account accepts TLS < 1.2

Severity: **Medium** Affected: 2 of 10 evaluated resources

Recommendation: Set minimum TLS version to 1.2.

Examples: stdevdata02, stdevdata04

Context: see 4.6 Storage in the infrastructure overview.

SEC-004: Key Vault purge protection disabled

Severity: **Medium** Affected: 1 of 4 evaluated resources

Recommendation: Enable purge protection to guard secrets and keys against malicious or accidental deletion.

Examples: kv-dev-00

Context: see 4.9 Security services in the infrastructure overview.

SEC-005: SQL server reachable from public networks

Severity: **Medium** Affected: 1 of 2 evaluated resources

Recommendation: Disable public network access and connect through Private Endpoints.

Examples: sql-dev-01

Context: see 4.7 Databases in the infrastructure overview.

6.2 Reliability — 75/100 (Fair)

REL-001: VM has no zone or availability set

Severity: **Medium** Affected: 15 of 24 evaluated resources

Recommendation: Deploy production VMs across availability zones or in a scale set / availability set.

Examples: vm-dev-000, vm-dev-001, vm-dev-002, vm-dev-003, vm-dev-004 ...

Context: see 4.4 Compute in the infrastructure overview.

REL-003: Storage account uses locally redundant replication

Severity: **Low** Affected: 3 of 10 evaluated resources

Recommendation: Use ZRS/GZRS for production data that needs zone or regional resilience.

Examples: stdevdata02, stprddata00, stprddata04

Context: see 4.6 Storage in the infrastructure overview.

REL-004: SQL database not zone redundant

Severity: **Low** Affected: 3 of 6 evaluated resources

Recommendation: Enable zone redundancy on business-critical databases.

Examples: sql-dev-01/db1, sql-dev-01/db2, sql-prd-01/db1

Context: see 4.7 Databases in the infrastructure overview.

6.3 Cost Optimization — 78/100 (Fair)

COST-003: VM stopped but not deallocated

Severity: **High** Affected: 4 of 24 evaluated resources

Recommendation: Deallocate stopped VMs (Stop from the portal/CLI) or schedule auto-shutdown.

Examples: vm-dev-006, vm-dev-009, vm-prd-000, vm-prd-001

Context: see 4.4 Compute in the infrastructure overview.

COST-001: Unattached managed disk

Severity: **Medium** Affected: 10 of 34 evaluated resources

Recommendation: Snapshot if required, then delete orphaned disks.

Examples: disk-dev-orphan-0, disk-dev-orphan-1, disk-dev-orphan-2, disk-dev-orphan-3, disk-prd-orphan-0 ...

Context: see 4.4 Compute in the infrastructure overview.

COST-005: Empty App Service plan

Severity: **Medium** Affected: 2 of 4 evaluated resources

Recommendation: Delete or consolidate App Service plans with no apps.

Examples: asp-dev-00, asp-dev-01

Context: see 4.8 Application hosting in the infrastructure overview.

COST-002: Unassociated public IP address

Severity: **Low** Affected: 2 of 6 evaluated resources

Recommendation: Release public IPs that are not attached to any resource.

Examples: pip-eastus-0, pip-westeuropa-0

Context: see 4.5 Networking in the infrastructure overview.

COST-004: Deallocated VM

Severity: **Low** Affected: 1 of 24 evaluated resources

Recommendation: Confirm the VM is still needed; delete it and its disks if not.

Examples: vm-prd-008

Context: see 4.4 Compute in the infrastructure overview.

6.4 Operational Excellence — 73/100 (Fair)

OPS-001: Previous-generation VM size

Severity: **Low** Affected: 5 of 24 evaluated resources

Recommendation: Resize to current generation (v5/v6) SKUs for better price-performance.

Examples: vm-dev-006, vm-prd-001, vm-prd-002, vm-prd-004, vm-prd-008

Context: see 4.4 Compute in the infrastructure overview.

OPS-002: Log Analytics retention below target

Severity: **Low** Affected: 2 of 2 evaluated resources

Recommendation: Increase retention or archive to meet audit and investigation requirements.

Examples: log-dev-01, log-prd-01

Context: see 4.10 Monitoring and management in the infrastructure overview.

6.5 Governance — 86/100 (Good)

GOV-003: Classic (ASM) resource in use

Severity: **High** Affected: 1 of 1 evaluated resources

Recommendation: Migrate classic resources to Azure Resource Manager before platform retirement.

Examples: legacy-cloudservice

Context: see 4.11 Governance and tagging in the infrastructure overview.

GOV-001: Required tags missing

Severity: **Medium** Affected: 34 of 152 evaluated resources

Recommendation: Enforce the tagging standard with Azure Policy (Require/Inherit a tag) and back-fill existing resources.

Examples: app-prd-01, asp-dev-00, asp-dev-01, disk-dev-orphan-0, disk-prd-orphan-3 ...

Context: see 4.11 Governance and tagging in the infrastructure overview.

GOV-002: Resource outside approved regions

Severity: **Medium** Affected: 8 of 152 evaluated resources

Recommendation: Apply the 'Allowed locations' policy and plan migration of out-of-policy resources.

Examples: disk-dev-orphan-2, disk-prd-orphan-3, nic-prd-012, sql-prd-01, stprddata00 ...

Context: see 4.3 Regional footprint in the infrastructure overview.

7. Remediation roadmap

0-30 days (quick wins & critical risk)

ID	Action	Severity	Pillar	Resources
SEC-001	Enable 'Secure transfer required' on the storage account.	High	Security	4
SEC-002	Set allowBlobPublicAccess to false and use SAS or Entra ID for access.	High	Security	4
COST-003	Deallocate stopped VMs (Stop from the portal/CLI) or schedule auto-shutdown.	High	Cost Optimization	4
SEC-006	Enable 'HTTPS Only' on the App Service.	High	Security	2
SEC-007	Remove Internet-sourced RDP/SSH rules; use Azure Bastion or Just-in-Time VM access.	High	Security	2
GOV-003	Migrate classic resources to Azure Resource Manager before platform retirement.	High	Governance	1
COST-001	Snapshot if required, then delete orphaned disks.	Medium	Cost Optimization	10
COST-005	Delete or consolidate App Service plans with no apps.	Medium	Cost Optimization	2
COST-002	Release public IPs that are not attached to any resource.	Low	Cost Optimization	2
COST-004	Confirm the VM is still needed; delete it and its disks if not.	Low	Cost Optimization	1

30-60 days

ID	Action	Severity	Pillar	Resources
GOV-001	Enforce the tagging standard with Azure Policy (Require/Inherit a tag) and back-fill existing resources.	Medium	Governance	34
REL-001	Deploy production VMs across availability zones or in a scale set / availability set.	Medium	Reliability	15
GOV-002	Apply the 'Allowed locations' policy and plan migration of out-of-policy resources.	Medium	Governance	8
SEC-003	Set minimum TLS version to 1.2.	Medium	Security	2
SEC-004	Enable purge protection to guard secrets and keys against malicious or accidental deletion.	Medium	Security	1
SEC-005	Disable public network access and connect through Private Endpoints.	Medium	Security	1

60-90 days

ID	Action	Severity	Pillar	Resources
OPS-001	Resize to current generation (v5/v6) SKUs for better price-performance.	Low	Operational Excellence	5
REL-003	Use ZRS/GZRS for production data that needs zone or regional resilience.	Low	Reliability	3
REL-004	Enable zone redundancy on business-critical databases.	Low	Reliability	3
OPS-002	Increase retention or archive to meet audit and investigation requirements.	Low	Operational Excellence	2

Appendix A. Detailed findings

Rule	Severity	Resource	Type	Resource group	Subscription	Detail
COST-003	High	vm-dev-006	Virtual machines	rg-dev-app03	Contoso-NonProduction	VM is stopped but still allocated (compute is billed)
COST-003	High	vm-dev-009	Virtual machines	rg-dev-app02	Contoso-NonProduction	VM is stopped but still allocated (compute is billed)
COST-003	High	vm-prd-000	Virtual machines	rg-prd-app01	Contoso-Production	VM is stopped but still allocated (compute is billed)
COST-003	High	vm-prd-001	Virtual machines	rg-prd-app02	Contoso-Production	VM is stopped but still allocated (compute is billed)
GOV-003	High	legacy-cloudservice	Classic cloud services	rg-legacy	Contoso-Connectivity	Classic (ASM) deployment model resource
SEC-001	High	stdevdata01	Storage accounts	rg-dev-data	Contoso-NonProduction	HTTP traffic is permitted
SEC-001	High	stdevdata03	Storage accounts	rg-dev-data	Contoso-NonProduction	HTTP traffic is permitted
SEC-001	High	stprddata03	Storage accounts	rg-prd-data	Contoso-Production	HTTP traffic is permitted
SEC-001	High	stprddata04	Storage accounts	rg-prd-data	Contoso-Production	HTTP traffic is permitted
SEC-002	High	stdevdata00	Storage accounts	rg-dev-data	Contoso-NonProduction	Anonymous blob access is allowed
SEC-002	High	stdevdata01	Storage accounts	rg-dev-data	Contoso-NonProduction	Anonymous blob access is allowed
SEC-002	High	stprddata02	Storage accounts	rg-prd-data	Contoso-Production	Anonymous blob access is allowed
SEC-002	High	stprddata04	Storage accounts	rg-prd-data	Contoso-Production	Anonymous blob access is allowed
SEC-006	High	app-dev-01	App Services	rg-dev-web	Contoso-NonProduction	HTTPS-only is not enforced
SEC-006	High	app-dev-02	App Services	rg-dev-web	Contoso-NonProduction	HTTPS-only is not enforced
SEC-007	High	nsg-mgmt-eastus	Network security groups	rg-hub-eastus	Contoso-Connectivity	Internet-exposed management ports: allow-rdp (3389)
SEC-007	High	nsg-mgmt-westeuropa	Network security groups	rg-hub-westeuropa	Contoso-Connectivity	Internet-exposed management ports: allow-rdp (3389)
COST-001	Medium	disk-dev-orphan-0	Managed disks	rg-dev-app01	Contoso-NonProduction	Unattached Premium_LRS disk, 128 GB
COST-001	Medium	disk-dev-orphan-1	Managed disks	rg-dev-app01	Contoso-NonProduction	Unattached Premium_LRS disk, 64 GB
COST-001	Medium	disk-dev-orphan-2	Managed disks	rg-dev-app01	Contoso-NonProduction	Unattached StandardSSD_LRS disk, 64 GB
COST-001	Medium	disk-dev-orphan-3	Managed disks	rg-dev-app01	Contoso-NonProduction	Unattached Premium_LRS disk, 64 GB
COST-001	Medium	disk-prd-orphan-0	Managed disks	rg-prd-app01	Contoso-Production	Unattached StandardSSD_LRS disk, 64 GB
COST-001	Medium	disk-prd-orphan-1	Managed disks	rg-prd-app01	Contoso-Production	Unattached Premium_LRS disk, 512 GB
COST-001	Medium	disk-prd-orphan-2	Managed disks	rg-prd-app01	Contoso-Production	Unattached Premium_LRS disk, 512 GB
COST-001	Medium	disk-prd-orphan-3	Managed disks	rg-prd-app01	Contoso-Production	Unattached Premium_LRS disk, 128 GB
COST-001	Medium	disk-prd-orphan-4	Managed disks	rg-prd-app01	Contoso-Production	Unattached StandardSSD_LRS disk, 256 GB
COST-001	Medium	disk-prd-orphan-5	Managed disks	rg-prd-app01	Contoso-Production	Unattached Premium_LRS disk, 64 GB
COST-005	Medium	asp-dev-00	App Service plans	rg-dev-web	Contoso-NonProduction	App Service plan hosts no apps
COST-005	Medium	asp-dev-01	App Service plans	rg-dev-web	Contoso-NonProduction	App Service plan hosts no apps
GOV-001	Medium	app-prd-01	App Services	rg-prd-web	Contoso-Production	Missing tag(s): environment, owner
GOV-001	Medium	asp-dev-00	App Service plans	rg-dev-web	Contoso-NonProduction	Missing tag(s): environment, owner
GOV-001	Medium	asp-dev-01	App Service plans	rg-dev-web	Contoso-NonProduction	Missing tag(s): environment, owner
GOV-001	Medium	disk-dev-orphan-0	Managed disks	rg-dev-app01	Contoso-NonProduction	Missing tag(s): costCenter
GOV-001	Medium	disk-prd-orphan-3	Managed disks	rg-prd-app01	Contoso-Production	Missing tag(s): costCenter
GOV-001	Medium	kv-prd-00	Key vaults	rg-prd-sec	Contoso-Production	Missing tag(s): environment, owner, costCenter

Rule	Severity	Resource	Type	Resource group	Subscription	Detail
GOV-001	Medium	kv-prd-01	Key vaults	rg-prd-sec	Contoso-Production	Missing tag(s): costCenter
GOV-001	Medium	nic-dev-002	Network interfaces	rg-dev-app03	Contoso-NonProduction	Missing tag(s): environment, owner, costCenter
GOV-001	Medium	nic-dev-004	Network interfaces	rg-dev-app01	Contoso-NonProduction	Missing tag(s): environment, owner, costCenter
GOV-001	Medium	nic-dev-005	Network interfaces	rg-dev-app02	Contoso-NonProduction	Missing tag(s): environment
GOV-001	Medium	nic-dev-006	Network interfaces	rg-dev-app03	Contoso-NonProduction	Missing tag(s): owner, costCenter
GOV-001	Medium	nic-prd-003	Network interfaces	rg-prd-app04	Contoso-Production	Missing tag(s): environment, costCenter
GOV-001	Medium	nic-prd-006	Network interfaces	rg-prd-app03	Contoso-Production	Missing tag(s): environment, owner
GOV-001	Medium	nsg-app-westeuropa	Network security groups	rg-hub-westeuropa	Contoso-Connectivity	Missing tag(s): environment, owner, costCenter
GOV-001	Medium	pip-eastus-0	Public IP addresses	rg-hub-eastus	Contoso-Connectivity	Missing tag(s): environment, owner, costCenter
GOV-001	Medium	pip-eastus-1	Public IP addresses	rg-hub-eastus	Contoso-Connectivity	Missing tag(s): owner, costCenter
GOV-001	Medium	pip-eastus-2	Public IP addresses	rg-hub-eastus	Contoso-Connectivity	Missing tag(s): environment, owner
GOV-001	Medium	sql-dev-01	SQL servers	rg-dev-data	Contoso-NonProduction	Missing tag(s): environment, costCenter
GOV-001	Medium	stdevdata02	Storage accounts	rg-dev-data	Contoso-NonProduction	Missing tag(s): environment, owner, costCenter
GOV-001	Medium	stprddata04	Storage accounts	rg-prd-data	Contoso-Production	Missing tag(s): environment, owner
GOV-001	Medium	vm-dev-003	Virtual machines	rg-dev-app04	Contoso-NonProduction	Missing tag(s): owner, costCenter
GOV-001	Medium	vm-dev-003_OsDisk	Managed disks	rg-dev-app04	Contoso-NonProduction	Missing tag(s): environment, owner, costCenter
GOV-001	Medium	vm-dev-006	Virtual machines	rg-dev-app03	Contoso-NonProduction	Missing tag(s): environment, owner, costCenter
GOV-001	Medium	vm-dev-008	Virtual machines	rg-dev-app01	Contoso-NonProduction	Missing tag(s): environment, owner, costCenter
GOV-001	Medium	vm-prd-000_OsDisk	Managed disks	rg-prd-app01	Contoso-Production	Missing tag(s): environment, owner, costCenter
GOV-001	Medium	vm-prd-001	Virtual machines	rg-prd-app02	Contoso-Production	Missing tag(s): environment, owner, costCenter
GOV-001	Medium	vm-prd-003_OsDisk	Managed disks	rg-prd-app04	Contoso-Production	Missing tag(s): environment, owner, costCenter
GOV-001	Medium	vm-prd-004	Virtual machines	rg-prd-app01	Contoso-Production	Missing tag(s): environment, owner
GOV-001	Medium	vm-prd-005_OsDisk	Managed disks	rg-prd-app02	Contoso-Production	Missing tag(s): environment, owner, costCenter
GOV-001	Medium	vm-prd-007_OsDisk	Managed disks	rg-prd-app04	Contoso-Production	Missing tag(s): environment, owner, costCenter
GOV-001	Medium	vm-prd-009	Virtual machines	rg-prd-app02	Contoso-Production	Missing tag(s): environment, owner, costCenter
GOV-001	Medium	vm-prd-009_OsDisk	Managed disks	rg-prd-app02	Contoso-Production	Missing tag(s): costCenter
GOV-001	Medium	vm-prd-010	Virtual machines	rg-prd-app03	Contoso-Production	Missing tag(s): environment, owner, costCenter
GOV-001	Medium	vm-prd-013_OsDisk	Managed disks	rg-prd-app02	Contoso-Production	Missing tag(s): environment, owner, costCenter
GOV-002	Medium	disk-dev-orphan-2	Managed disks	rg-dev-app01	Contoso-NonProduction	Deployed in 'centralindia', outside approved regions
GOV-002	Medium	disk-prd-orphan-3	Managed disks	rg-prd-app01	Contoso-Production	Deployed in 'centralindia', outside approved regions
GOV-002	Medium	nic-prd-012	Network interfaces	rg-prd-app01	Contoso-Production	Deployed in 'centralindia', outside approved regions
GOV-002	Medium	sql-prd-01	SQL servers	rg-prd-data	Contoso-Production	Deployed in 'centralindia', outside approved regions
GOV-002	Medium	stprddata00	Storage accounts	rg-prd-data	Contoso-Production	Deployed in 'centralindia', outside approved regions
GOV-002	Medium	stprddata02	Storage accounts	rg-prd-data	Contoso-Production	Deployed in 'centralindia', outside approved regions
GOV-002	Medium	vm-prd-012	Virtual machines	rg-prd-app01	Contoso-Production	Deployed in 'centralindia', outside approved regions
GOV-002	Medium	vm-prd-012_OsDisk	Managed disks	rg-prd-app01	Contoso-Production	Deployed in 'centralindia', outside approved regions
REL-001	Medium	vm-dev-000	Virtual machines	rg-dev-app01	Contoso-NonProduction	Not deployed to an availability zone, availability set or scale set

Rule	Severity	Resource	Type	Resource group	Subscription	Detail
REL-001	Medium	vm-dev-001	Virtual machines	rg-dev-app02	Contoso-NonProduction	Not deployed to an availability zone, availability set or scale set
REL-001	Medium	vm-dev-002	Virtual machines	rg-dev-app03	Contoso-NonProduction	Not deployed to an availability zone, availability set or scale set
REL-001	Medium	vm-dev-003	Virtual machines	rg-dev-app04	Contoso-NonProduction	Not deployed to an availability zone, availability set or scale set
REL-001	Medium	vm-dev-004	Virtual machines	rg-dev-app01	Contoso-NonProduction	Not deployed to an availability zone, availability set or scale set
REL-001	Medium	vm-dev-005	Virtual machines	rg-dev-app02	Contoso-NonProduction	Not deployed to an availability zone, availability set or scale set
REL-001	Medium	vm-dev-006	Virtual machines	rg-dev-app03	Contoso-NonProduction	Not deployed to an availability zone, availability set or scale set
REL-001	Medium	vm-dev-007	Virtual machines	rg-dev-app04	Contoso-NonProduction	Not deployed to an availability zone, availability set or scale set
REL-001	Medium	vm-dev-008	Virtual machines	rg-dev-app01	Contoso-NonProduction	Not deployed to an availability zone, availability set or scale set
REL-001	Medium	vm-dev-009	Virtual machines	rg-dev-app02	Contoso-NonProduction	Not deployed to an availability zone, availability set or scale set
REL-001	Medium	vm-prd-003	Virtual machines	rg-prd-app04	Contoso-Production	Not deployed to an availability zone, availability set or scale set
REL-001	Medium	vm-prd-004	Virtual machines	rg-prd-app01	Contoso-Production	Not deployed to an availability zone, availability set or scale set
REL-001	Medium	vm-prd-006	Virtual machines	rg-prd-app03	Contoso-Production	Not deployed to an availability zone, availability set or scale set
REL-001	Medium	vm-prd-007	Virtual machines	rg-prd-app04	Contoso-Production	Not deployed to an availability zone, availability set or scale set
REL-001	Medium	vm-prd-009	Virtual machines	rg-prd-app02	Contoso-Production	Not deployed to an availability zone, availability set or scale set
SEC-003	Medium	stdevdata02	Storage accounts	rg-dev-data	Contoso-NonProduction	Minimum TLS version is TLS1_0
SEC-003	Medium	stdevdata04	Storage accounts	rg-dev-data	Contoso-NonProduction	Minimum TLS version is TLS1_0
SEC-004	Medium	kv-dev-00	Key vaults	rg-dev-sec	Contoso-NonProduction	Purge protection is disabled
SEC-005	Medium	sql-dev-01	SQL servers	rg-dev-data	Contoso-NonProduction	Public network access is enabled
COST-002	Low	pip-eastus-0	Public IP addresses	rg-hub-eastus	Contoso-Connectivity	Public IP is not associated with any resource
COST-002	Low	pip-westeuropa-0	Public IP addresses	rg-hub-westeuropa	Contoso-Connectivity	Public IP is not associated with any resource
COST-004	Low	vm-prd-008	Virtual machines	rg-prd-app01	Contoso-Production	VM is deallocated; disks and IPs still incur cost
OPS-001	Low	vm-dev-006	Virtual machines	rg-dev-app03	Contoso-NonProduction	Previous-generation size Standard_A2_v2
OPS-001	Low	vm-prd-001	Virtual machines	rg-prd-app02	Contoso-Production	Previous-generation size Standard_DS2_v2
OPS-001	Low	vm-prd-002	Virtual machines	rg-prd-app03	Contoso-Production	Previous-generation size Standard_A2_v2
OPS-001	Low	vm-prd-004	Virtual machines	rg-prd-app01	Contoso-Production	Previous-generation size Standard_DS2_v2
OPS-001	Low	vm-prd-008	Virtual machines	rg-prd-app01	Contoso-Production	Previous-generation size Standard_A2_v2
OPS-002	Low	log-dev-01	Log Analytics workspaces	rg-dev-mon	Contoso-NonProduction	Retention is 30 days (target ≥ 90)
OPS-002	Low	log-prd-01	Log Analytics workspaces	rg-prd-mon	Contoso-Production	Retention is 30 days (target ≥ 90)
REL-003	Low	stdevdata02	Storage accounts	rg-dev-data	Contoso-NonProduction	Replication is Standard_LRS (single datacenter)
REL-003	Low	stprddata00	Storage accounts	rg-prd-data	Contoso-Production	Replication is Standard_LRS (single datacenter)
REL-003	Low	stprddata04	Storage accounts	rg-prd-data	Contoso-Production	Replication is Standard_LRS (single datacenter)

Rule	Severity	Resource	Type	Resource group	Subscription	Detail
REL-004	Low	sql-dev-01/db1	SQL databases	rg-dev-data	Contoso-NonProduction	Database is not zone redundant
REL-004	Low	sql-dev-01/db2	SQL databases	rg-dev-data	Contoso-NonProduction	Database is not zone redundant
REL-004	Low	sql-prd-01/db1	SQL databases	rg-prd-data	Contoso-Production	Database is not zone redundant

Appendix B. Resource inventory

Name	Type	Resource group	Location	Subscription	SKU
privatelink.blob.core.windows.net	privatednszones	rg-dns	global	Contoso-Connectivity	
privatelink.database.windows.net	privatednszones	rg-dns	global	Contoso-Connectivity	
afw-hub-eastus	Azure Firewalls	rg-hub-eastus	eastus	Contoso-Connectivity	AZFW_VNet
bas-hub-eastus	bastionhosts	rg-hub-eastus	eastus	Contoso-Connectivity	Standard
cn-hub-eastus-to-dc1	connections	rg-hub-eastus	eastus	Contoso-Connectivity	
lng-onprem-dc1	localnetworkgateways	rg-hub-eastus	eastus	Contoso-Connectivity	
nsg-app-eastus	Network security groups	rg-hub-eastus	eastus	Contoso-Connectivity	
nsg-mgmt-eastus	Network security groups	rg-hub-eastus	eastus	Contoso-Connectivity	
pip-eastus-0	Public IP addresses	rg-hub-eastus	eastus	Contoso-Connectivity	Standard
pip-eastus-1	Public IP addresses	rg-hub-eastus	eastus	Contoso-Connectivity	Standard
pip-eastus-2	Public IP addresses	rg-hub-eastus	eastus	Contoso-Connectivity	Standard
vgw-hub-eastus	virtualnetworkgateways	rg-hub-eastus	eastus	Contoso-Connectivity	VpnGw2AZ
vnet-hub-eastus	Virtual networks	rg-hub-eastus	eastus	Contoso-Connectivity	
afw-hub-westeuropa	Azure Firewalls	rg-hub-westeuropa	westeuropa	Contoso-Connectivity	AZFW_VNet
nsg-app-westeuropa	Network security groups	rg-hub-westeuropa	westeuropa	Contoso-Connectivity	
nsg-mgmt-westeuropa	Network security groups	rg-hub-westeuropa	westeuropa	Contoso-Connectivity	
pip-westeuropa-0	Public IP addresses	rg-hub-westeuropa	westeuropa	Contoso-Connectivity	Standard
pip-westeuropa-1	Public IP addresses	rg-hub-westeuropa	westeuropa	Contoso-Connectivity	Standard
pip-westeuropa-2	Public IP addresses	rg-hub-westeuropa	westeuropa	Contoso-Connectivity	Standard
vnet-hub-westeuropa	Virtual networks	rg-hub-westeuropa	westeuropa	Contoso-Connectivity	
legacy-cloudservice	Classic cloud services	rg-legacy	eastus	Contoso-Connectivity	
disk-dev-orphan-0	Managed disks	rg-dev-app01	westeuropa	Contoso-NonProduction	Premium_LRS
disk-dev-orphan-1	Managed disks	rg-dev-app01	westeuropa	Contoso-NonProduction	Premium_LRS
disk-dev-orphan-2	Managed disks	rg-dev-app01	centralindia	Contoso-NonProduction	StandardSSD_LRS
disk-dev-orphan-3	Managed disks	rg-dev-app01	eastus	Contoso-NonProduction	Premium_LRS
vm-dev-000_OsDisk	Managed disks	rg-dev-app01	eastus	Contoso-NonProduction	Premium_LRS
vm-dev-004_OsDisk	Managed disks	rg-dev-app01	eastus	Contoso-NonProduction	Premium_LRS
vm-dev-008_OsDisk	Managed disks	rg-dev-app01	eastus	Contoso-NonProduction	Premium_LRS
vm-dev-000	Virtual machines	rg-dev-app01	eastus	Contoso-NonProduction	
vm-dev-004	Virtual machines	rg-dev-app01	eastus	Contoso-NonProduction	
vm-dev-008	Virtual machines	rg-dev-app01	eastus	Contoso-NonProduction	

Name	Type	Resource group	Location	Subscription	SKU
nic-dev-000	Network interfaces	rg-dev-app01	eastus	Contoso-NonProduction	
nic-dev-004	Network interfaces	rg-dev-app01	eastus	Contoso-NonProduction	
nic-dev-008	Network interfaces	rg-dev-app01	eastus	Contoso-NonProduction	
vm-dev-001_OsDisk	Managed disks	rg-dev-app02	uksouth	Contoso-NonProduction	Premium_LRS
vm-dev-005_OsDisk	Managed disks	rg-dev-app02	uksouth	Contoso-NonProduction	Premium_LRS
vm-dev-009_OsDisk	Managed disks	rg-dev-app02	uksouth	Contoso-NonProduction	Premium_LRS
vm-dev-001	Virtual machines	rg-dev-app02	uksouth	Contoso-NonProduction	
vm-dev-005	Virtual machines	rg-dev-app02	uksouth	Contoso-NonProduction	
vm-dev-009	Virtual machines	rg-dev-app02	uksouth	Contoso-NonProduction	
nic-dev-001	Network interfaces	rg-dev-app02	uksouth	Contoso-NonProduction	
nic-dev-005	Network interfaces	rg-dev-app02	uksouth	Contoso-NonProduction	
nic-dev-009	Network interfaces	rg-dev-app02	uksouth	Contoso-NonProduction	
vm-dev-002_OsDisk	Managed disks	rg-dev-app03	eastus	Contoso-NonProduction	Premium_LRS
vm-dev-006_OsDisk	Managed disks	rg-dev-app03	uksouth	Contoso-NonProduction	Premium_LRS
vm-dev-002	Virtual machines	rg-dev-app03	eastus	Contoso-NonProduction	
vm-dev-006	Virtual machines	rg-dev-app03	uksouth	Contoso-NonProduction	
nic-dev-002	Network interfaces	rg-dev-app03	eastus	Contoso-NonProduction	
nic-dev-006	Network interfaces	rg-dev-app03	uksouth	Contoso-NonProduction	
vm-dev-003_OsDisk	Managed disks	rg-dev-app04	eastus	Contoso-NonProduction	Premium_LRS
vm-dev-007_OsDisk	Managed disks	rg-dev-app04	eastus	Contoso-NonProduction	Premium_LRS
vm-dev-003	Virtual machines	rg-dev-app04	eastus	Contoso-NonProduction	
vm-dev-007	Virtual machines	rg-dev-app04	eastus	Contoso-NonProduction	
nic-dev-003	Network interfaces	rg-dev-app04	eastus	Contoso-NonProduction	
nic-dev-007	Network interfaces	rg-dev-app04	eastus	Contoso-NonProduction	
sql-dev-01	SQL servers	rg-dev-data	uksouth	Contoso-NonProduction	
sql-dev-01/db0	SQL databases	rg-dev-data	westeurope	Contoso-NonProduction	Basic
sql-dev-01/db1	SQL databases	rg-dev-data	uksouth	Contoso-NonProduction	S1
sql-dev-01/db2	SQL databases	rg-dev-data	westeurope	Contoso-NonProduction	S1
stdevdata00	Storage accounts	rg-dev-data	westeurope	Contoso-NonProduction	Standard_GRS
stdevdata01	Storage accounts	rg-dev-data	eastus	Contoso-NonProduction	Standard_GRS
stdevdata02	Storage accounts	rg-dev-data	westeurope	Contoso-NonProduction	Standard_LRS
stdevdata03	Storage accounts	rg-dev-data	westeurope	Contoso-NonProduction	Standard_ZRS
stdevdata04	Storage accounts	rg-dev-data	westeurope	Contoso-NonProduction	Standard_GRS
log-dev-01	Log Analytics workspaces	rg-dev-mon	uksouth	Contoso-NonProduction	
vnet-dev-eastus	Virtual networks	rg-dev-network	eastus	Contoso-NonProduction	
vnet-dev-sandbox	Virtual networks	rg-dev-sandbox	uksouth	Contoso-NonProduction	
kv-dev-00	Key vaults	rg-dev-sec	westeurope	Contoso-NonProduction	
kv-dev-01	Key vaults	rg-dev-sec	eastus	Contoso-NonProduction	
asp-dev-00	App Service plans	rg-dev-web	eastus	Contoso-NonProduction	S1

Name	Type	Resource group	Location	Subscription	SKU
asp-dev-01	App Service plans	rg-dev-web	eastus	Contoso-NonProduction	B1
app-dev-00	App Services	rg-dev-web	westeurope	Contoso-NonProduction	
app-dev-01	App Services	rg-dev-web	eastus	Contoso-NonProduction	
app-dev-02	App Services	rg-dev-web	eastus	Contoso-NonProduction	
app-dev-03	App Services	rg-dev-web	uksouth	Contoso-NonProduction	
disk-prd-orphan-0	Managed disks	rg-prd-app01	uksouth	Contoso-Production	StandardSSD_LRS
disk-prd-orphan-1	Managed disks	rg-prd-app01	westeurope	Contoso-Production	Premium_LRS
disk-prd-orphan-2	Managed disks	rg-prd-app01	eastus	Contoso-Production	Premium_LRS
disk-prd-orphan-3	Managed disks	rg-prd-app01	centralindia	Contoso-Production	Premium_LRS
disk-prd-orphan-4	Managed disks	rg-prd-app01	eastus	Contoso-Production	StandardSSD_LRS
disk-prd-orphan-5	Managed disks	rg-prd-app01	westeurope	Contoso-Production	Premium_LRS
vm-prd-000_OsDisk	Managed disks	rg-prd-app01	westeurope	Contoso-Production	Premium_LRS
vm-prd-004_OsDisk	Managed disks	rg-prd-app01	westeurope	Contoso-Production	Premium_LRS
vm-prd-008_OsDisk	Managed disks	rg-prd-app01	uksouth	Contoso-Production	Premium_LRS
vm-prd-012_OsDisk	Managed disks	rg-prd-app01	centralindia	Contoso-Production	Premium_LRS
vm-prd-000	Virtual machines	rg-prd-app01	westeurope	Contoso-Production	
vm-prd-004	Virtual machines	rg-prd-app01	westeurope	Contoso-Production	
vm-prd-008	Virtual machines	rg-prd-app01	uksouth	Contoso-Production	
vm-prd-012	Virtual machines	rg-prd-app01	centralindia	Contoso-Production	
lbi-prd-app	Load balancers	rg-prd-app01	eastus	Contoso-Production	Standard
nic-prd-000	Network interfaces	rg-prd-app01	westeurope	Contoso-Production	
nic-prd-004	Network interfaces	rg-prd-app01	westeurope	Contoso-Production	
nic-prd-008	Network interfaces	rg-prd-app01	uksouth	Contoso-Production	
nic-prd-012	Network interfaces	rg-prd-app01	centralindia	Contoso-Production	
vm-prd-001_OsDisk	Managed disks	rg-prd-app02	eastus	Contoso-Production	Premium_LRS
vm-prd-005_OsDisk	Managed disks	rg-prd-app02	eastus	Contoso-Production	Premium_LRS
vm-prd-009_OsDisk	Managed disks	rg-prd-app02	eastus	Contoso-Production	Premium_LRS
vm-prd-013_OsDisk	Managed disks	rg-prd-app02	westeurope	Contoso-Production	Premium_LRS
vm-prd-001	Virtual machines	rg-prd-app02	eastus	Contoso-Production	
vm-prd-005	Virtual machines	rg-prd-app02	eastus	Contoso-Production	
vm-prd-009	Virtual machines	rg-prd-app02	eastus	Contoso-Production	
vm-prd-013	Virtual machines	rg-prd-app02	westeurope	Contoso-Production	
nic-prd-001	Network interfaces	rg-prd-app02	eastus	Contoso-Production	
nic-prd-005	Network interfaces	rg-prd-app02	eastus	Contoso-Production	
nic-prd-009	Network interfaces	rg-prd-app02	eastus	Contoso-Production	
nic-prd-013	Network interfaces	rg-prd-app02	westeurope	Contoso-Production	
vm-prd-002_OsDisk	Managed disks	rg-prd-app03	uksouth	Contoso-Production	Premium_LRS
vm-prd-006_OsDisk	Managed disks	rg-prd-app03	eastus	Contoso-Production	Premium_LRS
vm-prd-010_OsDisk	Managed disks	rg-prd-app03	westeurope	Contoso-Production	Premium_LRS
vm-prd-002	Virtual machines	rg-prd-app03	uksouth	Contoso-Production	

Name	Type	Resource group	Location	Subscription	SKU
vm-prd-006	Virtual machines	rg-prd-app03	eastus	Contoso-Production	
vm-prd-010	Virtual machines	rg-prd-app03	westeurope	Contoso-Production	
nic-prd-002	Network interfaces	rg-prd-app03	uksouth	Contoso-Production	
nic-prd-006	Network interfaces	rg-prd-app03	eastus	Contoso-Production	
nic-prd-010	Network interfaces	rg-prd-app03	westeurope	Contoso-Production	
vm-prd-003_OsDisk	Managed disks	rg-prd-app04	westeurope	Contoso-Production	Premium_LRS
vm-prd-007_OsDisk	Managed disks	rg-prd-app04	eastus	Contoso-Production	Premium_LRS
vm-prd-011_OsDisk	Managed disks	rg-prd-app04	uksouth	Contoso-Production	Premium_LRS
vm-prd-003	Virtual machines	rg-prd-app04	westeurope	Contoso-Production	
vm-prd-007	Virtual machines	rg-prd-app04	eastus	Contoso-Production	
vm-prd-011	Virtual machines	rg-prd-app04	uksouth	Contoso-Production	
nic-prd-003	Network interfaces	rg-prd-app04	westeurope	Contoso-Production	
nic-prd-007	Network interfaces	rg-prd-app04	eastus	Contoso-Production	
nic-prd-011	Network interfaces	rg-prd-app04	uksouth	Contoso-Production	
sql-prd-01	SQL servers	rg-prd-data	centralindia	Contoso-Production	
sql-prd-01/db0	SQL databases	rg-prd-data	eastus	Contoso-Production	GP_Gen5_2
sql-prd-01/db1	SQL databases	rg-prd-data	eastus	Contoso-Production	S1
sql-prd-01/db2	SQL databases	rg-prd-data	eastus	Contoso-Production	Basic
stprddata00	Storage accounts	rg-prd-data	centralindia	Contoso-Production	Standard_LRS
stprddata01	Storage accounts	rg-prd-data	westeurope	Contoso-Production	Standard_GRS
stprddata02	Storage accounts	rg-prd-data	centralindia	Contoso-Production	Standard_ZRS
stprddata03	Storage accounts	rg-prd-data	eastus	Contoso-Production	Standard_ZRS
stprddata04	Storage accounts	rg-prd-data	eastus	Contoso-Production	Standard_LRS
log-prd-01	Log Analytics workspaces	rg-prd-mon	eastus	Contoso-Production	
nsg-prd-app	Network security groups	rg-prd-network	eastus	Contoso-Production	
nsg-prd-data	Network security groups	rg-prd-network	eastus	Contoso-Production	
nsg-prd-web	Network security groups	rg-prd-network	eastus	Contoso-Production	
pe-sql-prd-01	privateendpoints	rg-prd-network	eastus	Contoso-Production	
pe-stprddata00-blob	privateendpoints	rg-prd-network	eastus	Contoso-Production	
rt-prd-eastus	routetables	rg-prd-network	eastus	Contoso-Production	
rt-prd-weu	routetables	rg-prd-network	westeurope	Contoso-Production	
vnet-prd-eastus	Virtual networks	rg-prd-network	eastus	Contoso-Production	
vnet-prd-westeurope	Virtual networks	rg-prd-network	westeurope	Contoso-Production	
kv-prd-00	Key vaults	rg-prd-sec	eastus	Contoso-Production	
kv-prd-01	Key vaults	rg-prd-sec	westeurope	Contoso-Production	
agw-prd-eastus	Application gateways	rg-prd-web	eastus	Contoso-Production	
asp-prd-00	App Service plans	rg-prd-web	westeurope	Contoso-Production	P1v3
asp-prd-01	App Service plans	rg-prd-web	westeurope	Contoso-Production	P1v3

Name	Type	Resource group	Location	Subscription	SKU
app-prd-00	App Services	rg-prd-web	eastus	Contoso-Production	
app-prd-01	App Services	rg-prd-web	eastus	Contoso-Production	
app-prd-02	App Services	rg-prd-web	westeurope	Contoso-Production	
app-prd-03	App Services	rg-prd-web	eastus	Contoso-Production	